INDUSTRIAL HAZARDS

AND SAFETY

DEFINITIONS:

- HAZARD: A substance that causes an injury in a given environment or situation.
- INDUSTRIAL HAZARD: Any condition produced by industries that may cause injury or death to personnel or loss of product or property.
- SAFETY: Freedom from the occurrence of risk or injury or loss.
- INDUSTRIAL SAFETY: Protection of workers from the danger of industrial accidents.
- **Toxic and corrosive chemicals, fire, explosions and personnel falling-causes of accidents.**

CAUSES OF ACCIDENTS:

- Human failure:
  - Operating without clearance, failure to respond warning
  - Working at an unsafe speed
  - Using equipment unsafely
  - Making safety devices un-operative
  - Unsafe loading, placing, mixing
  - Distracting, teasing, abusing

- Machine failure:
  - Inadequately guarded, height, strength etc.
  - Unsafely designed machines, tools etc.
  - Defective, rough, sharp, slippery, decayed machines
  - Unsafely arranged, congestion, blocked exits.
  - Unsafely clothed, no goggles, glares or masks, high heels etc.

- Types of Hazards:
  - Electrical hazards
  - Chemical hazards
  - Gas hazards
  - Dust explosion
  - Fire and Explosion hazards.

CHEMICAL HAZARDS

- Reasons like:
  - Wrong raw materials
  - Raw materials containing trace impurities
  - Changed operating conditions
  - Unanticipated time delays
  - Equipment failure
  - Wrong materials of construction

Management of over exposure to chemicals:

1. Removal from exposure
2. Resuscitation: restoration of life
3. Decontamination
4. Symptomatic treatment: convulsions, bronchospasms, dehydration, and arrhythmias.

What are the effects???

- From irritation to death
- Severe burns by destroying living tissues- dehydration, digestion, oxidation
- Eyes, nervous system, nose, mouth and throat, pulmonary system, CVS, Breast, digestive system, genitourinary system, skin

- Dermatitis reactions to synthetic drugs- Acridines and phenothiazine compounds
- While grinding vegetable drugs dust evolved causes irritation- capsicum, podophyllum

GAS HAZARDS

- Flammable liquids:

Vaporization causes air pollution

Ignition of vapor causing fire accidents and explosions

Further spreading leads to loss of life and property

- Location, storage and handling:

The properties of liquids are considered to design suitable methods for storage and handling

1. Flash point
2. Apparent ignition temperature
3. Explosive range
4. Vapour density

- Stored in isolated places.
- Using tanks constructed above ground and open place
- Methods of handling include pumping, gravity flow and compressed air.

- Compressed gases: filled in cylinders and transported to the place of use and the precautions are:
  - Cylinders should not be dropped or struck.
  - Safety devices should not be dropped.
  - Valves should be of high quality.
  - Protected against weather changes, especially excessive temperature.
  - Cylinders should bear a standard label indicating the kind of gas- inflammable, corrosive, or inert.
  - Full cylinders should be separated from empty cylinders.

DUST EXPLOSION

- The term Dust is used if the Max particle size of the solids is 500 mm in the mixture.
- During grinding operations, stone or iron pieces results in sparks finally leads to explosion of combustible materials.
- Dust of starch, dextrin and some organic materials are dangerous.
- Mixture of combustible dust burns or oxidizes in presence of air.
- Upper and lower limits of air and dust are given for the explosion to takes place.

Methods to exclude dust explosion

- Avoiding the development of explosive mixtures
- Replacing the atmospheric oxygen with inert gas, working in a vacuum, or using inert dust-rock *salt and sod sulfate*
- Preventing the occurrence of an effective ignition source- *welding, smoking, cutting, mechanically generated sparks*

CONTROL OF DUST EXPLOSION

Can be controlled at the source of generation and the methods include

1. Filtration
2. Inertial separation
3. Electrostatic precipitation

Fire and Explosion Hazards

What is the source?

- Spontaneous combustion
- Non-explosion-proof electrical equipment

Fire occurs in the industry more frequently than an explosion, or toxic release, but the loss in terms of life is less…

Control Fire & Explosion

- Building construction
- Exit points
- Fire alarm equipment
- Fire Extinguishers
- Sprinkler systems
- Hydrants
- Safety and Fire protection organization

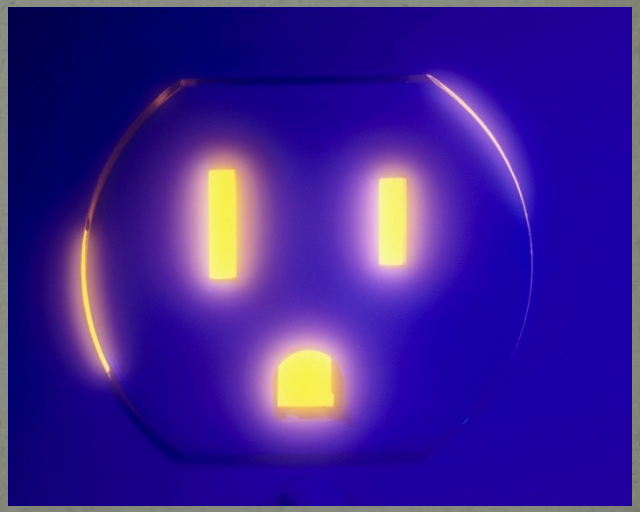
Electrical Hazards



Why know about it?

Because failing to take the necessary precautions can lead to:

- injury or death
- fire or property damage



What are the types of injuries?

- Burns
- Shocks
- Falls

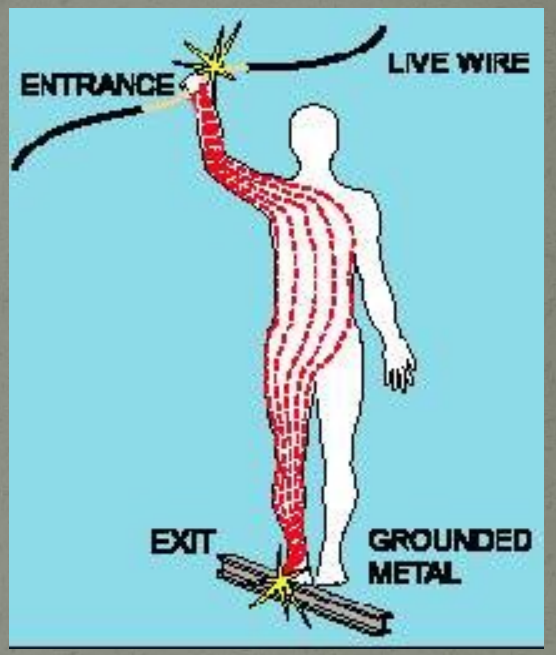


KEEP SAFE

KNOW HOW TO CONTROL ELECTRICAL HAZARDS

Shock Severity

- Severity of the shock depends on:
  - Path of current through the body
  - Amount of current flowing through the body (amps)
  - Duration of the shocking current through the body,
- LOW VOLTAGE DOES NOT MEAN LOW HAZARD



Electrical Injuries

There are three direct and two indirect types of electrical injuries:

- Direct:
  - Death due to electrical shock
  - Electrical shock
  - Burns
- Indirect:
  - Falls
  - Fire



What are the safety priorities?

Electrical hazards exist in almost every workplace. Common causes of electrocution are:

- making contact with overhead wires
- undertaking maintenance on live equipment
- working with damaged electrical equipment, such as extension leads, plugs and sockets
- using equipment affected by rain or water ingress

How do you respond to electrical incidents?

If you come across a person receiving an electric shock:

- if possible, disconnect the electrical supply (switch?)
- assess the situation – never put yourself at risk
- take precautions to protect yourself and anyone else in the vicinity
- apply the first aid principles
- assess the injuries and move the casualty to a safe area if required
- administer first aid if trained
- seek urgent medical attention



Do you want to be a victim?

You could be the victim if you:

- don't follow proper procedures around electricity
- use electrical equipment improperly
- use faulty electrical equipment

What are the levels of effect of current?

AC current (mA)	Effect on human body
1	Slight tingling sensation
2-9	Small shock
10-24	Muscles contract causing you to freeze
25-74	Respiratory muscles can become paralysed; pain; exit burns often visible
75-300	Usually fatal; ventricular fibrillation; entry & exit wounds visible
>300	Death almost certain; if survive will have badly burnt organs and probably require amputations



What should you do in an electrical emergency?

For low voltage electricity >50 V AC and 110 V DC

- remove the source of electricity supply
- call the emergency number on site

For high voltage electricity >1000 V

- call the emergency number for your site
- don't go near the casualty
- don't touch the casualty or try to free them with anything

Should you report electrical incidents?

- Electricity is invisible – this in itself makes it dangerous
- It has great potential to seriously injure or kill
- The company has a duty of care to its employees and contractors
- Everyone is exposed to electrical hazards, not just electricians
- Report all electrical shocks and near misses

All employees can be exposed to electrical Hazards. They should receive electrical hazard Training at the commencement of their Employment and REGULAR REFRESHER TRAINING.

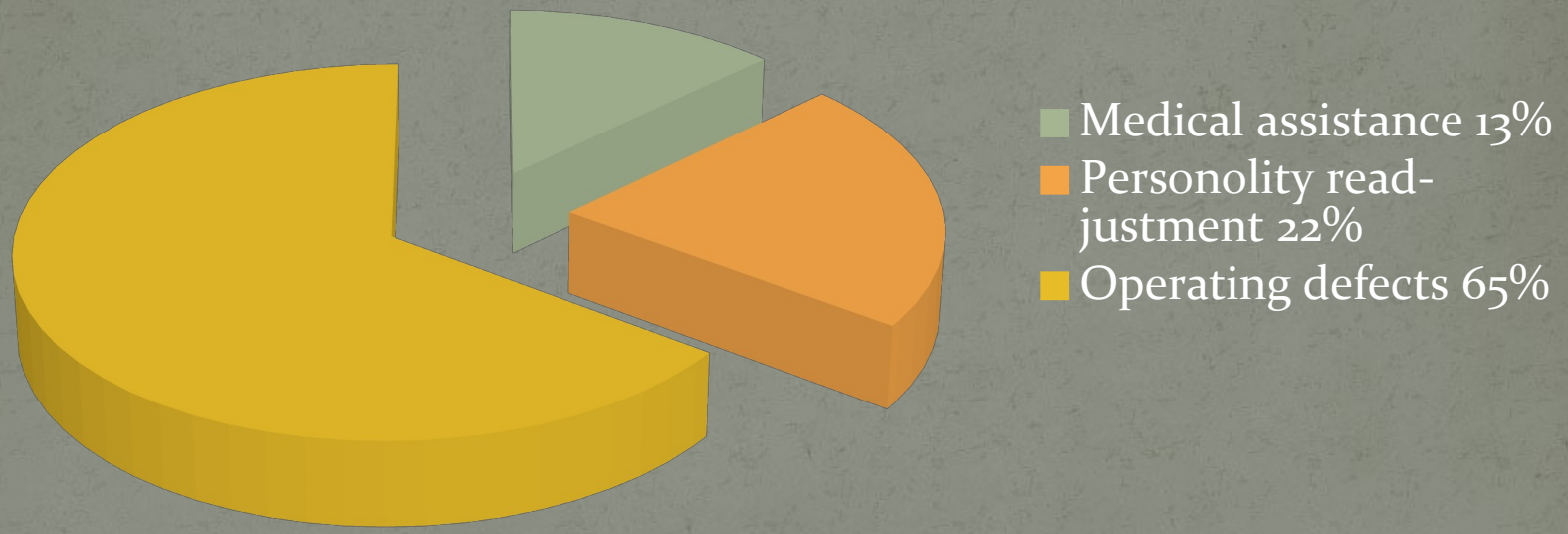
Can you protect yourself from electricity?

- Don't wear metal objects
- Turn power off
- Wear appropriate clothing
- Don't touch live parts
- Don't install or repair electrical equipment
- Use qualified personnel
- Clean and dry leads and plugs before use



Accident prevention & reduction

Treatment



SAFETY MANAGEMENT

- Types of accidents:
  - Major results in death or prolonged disability.
  - Minor scratch or a cut that doesn't seriously disable the worker.

Advantages of safety management:

1. Cost saving: Direct costs-compensation medical expenses

Indirect costs to 4 times higher include Loss due to downtime operations, slowed-up production rate, spoiled materials, labor time, charges for cleaning, and damage of equipment.

2. Increased productivity: employees can devote more time in improving the quality and quantity of output.
3. Moral:boosts morality of the worker and his family.
4. Legal: laws covering occupational health and safety and the penalties for noncompliance are severe.

Special safety protection equipment:

- Head and Eyes: goggles, helmets, masks and hats.
- Hands, arms, legs and feet: rubber or canvas gloves, rubber boots, aprons, shoes and other clothing.
- Prevention of breathing of poisonous gases: respiration protective devices.

Safety programme

- Safety programme deals with the prevention of accidents and minimizing the resulting loss or damage to persons and property.
- Principles of safety programme:
  - To identify potential hazards, provide safety equipment and facilities for taking prompt action.
  - To develop safety policies that continuously monitor safety performance and ensure that corrective action is taken when necessary.

- Extent and remedies for accidents:

  Based on the causes the supervisor should apply the art of supervision to control worker's performance to prevent accidents.

  Human failure and machine failure.

- safety training:
  - Systematic training of employees
  - Safety measures should be kept in mind while planning a layout and construction.
  - Workers must be provided with safety tools.
  - Safety posters with slogans at workplaces.
  - National Safety Council.

- To make supervision fully accountable for safety performance in the working areas.
- To train in safe methods of working and provide education and guidance on prevention of hazards.

ELEMENTS OF THE SAFETY PROGRAMME:

- Strategic choice
- Safety policy
- Organization for safety
- Extent and remedies for accidents
- Training in safety
- Safety program evaluation

- Strategic choice: To make decisions regarding safety of workers.
  As a marketing tool.
- Safety policy: must contain a declaration of 4 fundamental points-

i. Safety of employees and the public is paramount importance.
ii. Safety will take precedence over expediency.
iii. In the development and implementation of safety procedures every effort will be made to involve all managers, supervisors and employees.
iv. Safety legislation will be complied in letter and the spirit of the law.

- safety organization:

Constitution of an organization- Safety committee.

composed of employees.

serves as advisory capacities

responsible for – reviewing safety procedures.

-making recommendations for eliminating health hazards.

-investigating accidents

-fielding safety-related complaints from employees.

- Safety program evaluation:
  - Qualitatively: changes in the behavior and attitudes of the employees, changes in the injury-producing conditions.
  - Quantitatively: incident ratio, severity ratio, and frequency ratio.